

Multiple Choice

1. **Answer:** B
Options: A) Magnetic susceptibility; B) Hall coefficient; C) Electrical resistivity; D) Thermal conductivity
Note: Correct option: B - Hall coefficient
2. **Answer:** A
Options: A) 5; B) 4; C) 3; D) 2
Note: Correct option: A - 5
3. **Answer:** B
Options: A) 0.50 c; B) 0.60 c; C) 0.70 c; D) 0.80 c
Note: Correct option: B - 0.60 c
4. **Answer:** A
Options: A) Planck's constant; B) Boltzmann's constant; C) The Rydberg constant; D) The speed of light
Note: Correct option: A - Planck's constant
5. **Answer:** A
Options: A) 4; B) 5; C) 6; D) 20
Note: Correct option: A - 4

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '450 m'.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Increases by a factor of 81.'

Long Form Response

11. **Answer:** 3/2. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 10. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key